SIMATS ENGINEERING

Saveetha Institute of Medical And Technical Sciences

Department of Computer Science and Engineering

Course Code & Name: CSA06 - Design and Analysis of Algorithms

Assessment: CO1 - Assessment Tool 1: IT Application-Based Problem Solving - Scalability Audit of an E-Commerce Platform

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Topic: Scalability Audit of an E-Commerce Platform ("ShopSphere")

COMPREHENSIVE SOLUTIONS & SCALABILITY AUDIT REPORT

1. Executive Summary & Theoretical Foundations of Algorithm Analysis

As ShopSphere scales from n = 100,000 to n = 1,000,000 products, evaluating the computational complexity of back-end modules becomes mission-critical [cite: 1, 2]. Algorithm analysis relies on asymptotic notations (Big-O, Omega, Theta) to characterize resource growth rates independently of hardware specifics [cite: 1, 2]. • **Big-O Notation O(g(n)):** Represents an upper bound, defining the worst-case asymptotic growth [cite: 2].
• **Omega Notation Omega(g(n)):** Represents a lower bound, defining the best-case behavior [cite: 2].
• **Theta Notation Theta(g(n)):** Represents a tight bound, where a function is bounded both above and below asymptotically [cite: 2].
This audit investigates five core backend modules to identify scalability bottlenecks, apply rigorous mathematical techniques (Substitution, Master Theorem, and asymptotic proofs), and recommend robust architectural redesigns [cite: 1, 2].

(a) Understand the Problem

For each backend module of ShopSphere, the input size n, basic operation, location, growth function, and detailed justification are set out below [cite: 1, 2]:

Module	Input Size (n)	Basic Operation	Location in Code	Growth/Detailed Justification
M1: Duplicate Detector	Catalogue products count (n) [cite: 1]	Comparison of product titles & sellers [cite: 1]	IF condition in inner loop [cite: 1]	Quadratic ~ n^2 / 2 operations. Nested loops compare every possible pair [cite: 1, 2].
M2: API Rate Limiter	Window size parameter (n) [cite: 1]	Audit call: Audit(Request[j]) [cite: 1]	Inside inner FOR loop [cite: 1]	Linear O(n). Inner loop executes 1 + 2 + 4 + ... < n times in total [cite: 1, 2].
M3: Product Search	Catalogue products count (n) [cite: 1, 2]	Key comparison with middle element [cite: 1, 2]	Search loop/recursion [cite: 1, 2]	S1: Linear O(n) sequential scan [cite: 1, 2]. S2: Logarithmic O(log n) binary search split [cite: 1, 2].
M4: Recommendation	Customer-product data size (n) [cite: 2]	Merging partial results / key comparisons [cite: 2]	Merge step across levels [cite: 2]	Current: O(n^2) via T(n)=4T(n/2)+n [cite: 2]. Redesign: O(n log n) via T(n)=2T(n/2)+n [cite: 2].

M5: Sales Analytics	Sales records count (n) [cite: 2]	Record comparison / revenue sorting [cite: 2]	Sorting step across log n levels [cite: 2]	Analytics: O(n log^2 n) [cite: 2]. Report Gen: O(n^2) via 3n^2 term [cite: 2].

(b) Analyse the Iterative Modules

Analytical Derivations & Calculations:

• **Module M1 Exact Comparison Derivation C(n):**

1. C(n) = sum_{i=0}^{n-2} sum_{j=i+1}^{n-1} 1 [cite: 1, 2]
2. Inner loop iterations for a given i: (n - 1) - (i + 1) + 1 = n - 1 - i [cite: 1, 2]
3. Summing over outer loop: C(n) = (n - 1) + (n - 2) + ... + 2 + 1 [cite: 1, 2]
4. Closed-form summation: C(n) = [n(n - 1)] / 2 = (n^2 - n) / 2 [cite: 1, 2]

• **Module M2 Total Audit Operations A(n):**

1. Outer loop variable i takes values: 1, 2, 4, 8, ..., 2^k where 2^k < n [cite: 1, 2]
2. Inner loop executes i times per outer iteration [cite: 1, 2]
3. Total operations A(n) = sum_{k=0}^{m-1} 2^k where m = ceil(log2 n) [cite: 1, 2]
4. Geometric series summation: A(n) = 2^m - 1 = 2^(ceil(log2 n)) - 1 [cite: 1, 2]
5. Verification for n = 1,024 = 2^10: A(1,024) = 2^10 - 1 = 1,023 Audit operations [cite: 1, 2].

Explanations & Interpretation:

• **Module M1 Execution Pattern:** The nested loops in M1 compare every product with every subsequent product, resulting in a quadratic growth rate of Theta(n^2) and O(1) auxiliary space [cite: 1, 2].

• **Module M2 Rate Limiter Analysis:** The team member's claim that M2 is Theta(n log n) is incorrect because the inner loop size grows exponentially (1, 2, 4, ...) rather than running n times every step [cite: 1, 2]. The tight asymptotic bound is linear Theta(n) [cite: 1, 2].

(c) Solve a Recurrence by Substitution

Analytical Derivations & Calculations:

• **Recurrence Relation:** T(n) = T(n/2) + 1 for n > 1, with T(1) = 1 [cite: 2].

• **Backward Substitution Steps:**

1. T(n) = T(n/2) + 1 [cite: 2]
2. T(n/2) = T(n/4) + 1 => T(n) = [T(n/4) + 1] + 1 = T(n/4) + 2 [cite: 2]
3. T(n/4) = T(n/8) + 1 => T(n) = [T(n/8) + 1] + 2 = T(n/8) + 3 [cite: 2]
4. General pattern after k steps: T(n) = T(n / 2^k) + k [cite: 2]

• **Base Case Resolution:**

1. Set n / 2^k = 1 => 2^k = n => k = log2 n [cite: 2]
2. Substitute k back: T(n) = T(1) + log2 n = 1 + log2 n [cite: 2]
3. Exact Closed Form: T(n) = log2 n + 1 [cite: 2].

Explanations & Interpretation:

• **Efficiency & Complexity:** Product search version S2 exhibits logarithmic time complexity Theta(log n) [cite: 2].

• **Space Complexity:** The iterative implementation requires Theta(1) auxiliary space, whereas the recursive implementation requires Theta(log n) auxiliary space due to the call stack depth of log2 n frames [cite: 2].

(d) Apply the Master Theorem

Analytical Derivations & Calculations:

• **Module M4 Current Design: T(n) = 4T(n/2) + n** [cite: 2]

- Parameters: a = 4, b = 2, f(n) = n [cite: 2]. Critical exponent n^(log_b a) = n^2 [cite: 2].
- Master Theorem Case 1 applies since f(n) = O(n^(2-e)) [cite: 2]. Complexity: Theta(n^2) [cite: 2].
- Doubling data (n -> 2n): T(2n) = (2n)^2 = 4n^2 = 4T(n) [cite: 2].

• **Module M4 Proposed Redesign: T(n) = 2T(n/2) + n** [cite: 2]

- Parameters: a = 2, b = 2, f(n) = n [cite: 2]. Critical exponent n^(log_b a) = n [cite: 2].
- Master Theorem Case 2 applies since f(n) = Theta(n) [cite: 2]. Complexity: Theta(n log n) [cite: 2].
- Doubling data (n -> 2n): T(2n) = 2n log(2n) approx 2T(n) [cite: 2].

• **Module M5 Sales Analytics Engine: T(n) = 2T(n/2) + n log n** [cite: 2]

- Parameters: a = 2, b = 2, f(n) = n log n. Critical exponent n = n [cite: 2].
- Standard 3-case Master Theorem does not apply directly [cite: 2].
- Extended Master Theorem (Case 2 Generalized) / Recursion Tree: Total work = sum_{j=0}^{log2 n - 1} n(log n - j) = Theta(n log^2 n) [cite: 2].

Explanations & Interpretation:

• **M4 Comparison:** Removing replication in M4 shifts the complexity from quadratic Theta(n^2) to log-linear Theta(n log n), transforming the platform's scalability when data doubles [cite: 2].

• **M5 Analytics Behavior:** The analytics engine recurrence yields Theta(n log^2 n), reflecting recursive partitioning combined with level-by-level sorting [cite: 2].

(e) Analyse Best, Average and Worst Cases

Version	Best Case Condition	Best	Worst Case Condition	Worst	Average Case Formula
S1: Linear Search	Target is at 1st position [cite: 2]	1 [cite: 2]	Target at last position or absent [cite: 2]	n [cite: 2]	C_avg = p(n+1)/2 + (1-p)n [cite: 2]
S2: Binary Search	Target is middle element [cite: 2]	1 [cite: 2]	Target at leaf or absent [cite: 2]	floor(log2 n) + 1 [cite: 2]	= log2 n - 1 (for hit) [cite: 2]

Analytical Derivations & Calculations:

• **S1 Average Case Derivation:**

1. Probability target exists = p [cite: 2]. Expected comparisons when present = (n + 1) / 2 [cite: 2].
2. Probability target is absent = 1 - p [cite: 2]. Comparisons when absent = n [cite: 2].
3. C_avg(n) = p * [(n + 1) / 2] + (1 - p) * n [cite: 2]

• **Evaluation for n = 1,000,000 and p = 0.5:**

C_avg(1,000,000) = 0.5 * [(1,000,000 + 1) / 2] + (1 - 0.5) * 1,000,000 [cite: 2]
= 0.5 * 500,000.5 + 500,000 = 250,000.25 + 500,000 = 750,000.25 comparisons [cite: 2].

Explanations & Interpretation:

• **Search Behavior:** Sequential search S1 scales linearly with catalogue size in the worst and average cases, whereas binary search S2 restricts searches to logarithmic bounds Theta(log n) [cite: 2].

(f) Prove an Asymptotic Bound

Analytical Derivations & Calculations:

• **Report Generator Cost Function:** f(n) = 3n^2 + 10n log2 n + 50 [cite: 2].

• **Part 1: Proof that f(n) = Theta(n^2):**

1. Lower Bound: For all n >= 1, 10n log2 n > 0 and 50 > 0, so f(n) > 3n^2 (c1 = 3, n0 = 1) [cite: 2].
2. Upper Bound: For all n >= 10, log2 n <= n => 10n log2 n <= 10n^2 and 50 <= 1n^2 [cite: 2].
3. f(n) <= 3n^2 + 10n^2 + 1n^2 = 14n^2 (c2 = 14, n0 = 10) [cite: 2].
4. Thus 3n^2 <= f(n) <= 14n^2 for all n >= 10, proving f(n) = Theta(n^2) [cite: 2].

• **Part 2: Proof that f(n) is NOT O(n):**

1. Assume by contradiction that f(n) = O(n), so 3n^2 + 10n log2 n + 50 <= c * n for some constant c [cite: 2].
2. Divide by n: 3n + 10 log2 n + (50/n) <= c [cite: 2].
3. As n -> infinity, the left side approaches infinity, contradicting the upper bound constant c [cite: 2].

Explanations & Interpretation:

• **Asymptotic Strictness:** The formal proof confirms that the report generator's quadratic growth is strictly bounded by Theta(n^2) and cannot be linear O(n) [cite: 2].

(g) Scalability Decision & Recommendations

Server Speed: 10^8 ops/sec. Nightly Window (10 min = 600 s): Limit = 6 x 10^10 ops. Target n = 1,000,000 [cite: 2].

Module	Formula	Operations (n=10^6)	Time (s)	10-Min Limit?	Status
M1: Duplicate Detector	(n^2 - n) / 2 [cite: 2]	4.999995 x 10^11 [cite: 2]	4,999.99 s (~83.3 min) [cite: 2]	NO (Exceeds by 8.3x) [cite: 2]	CRITICAL FAIL [cite: 2]
M2: API Rate Limiter	2^(ceil(log2 n)) - 1 [cite: 2]	1,048,575 [cite: 2]	0.0105 s [cite: 2]	YES [cite: 2]	PASS [cite: 2]
M3 (S2): Product Search	log2 n + 1 [cite: 2]	21 [cite: 2]	0.00000021 s [cite: 2]	YES [cite: 2]	PASS [cite: 2]
M4 Current Recommendation	n^2 [cite: 2]	1.00 x 10^12 [cite: 2]	10,000.0 s (~166.7 min) [cite: 2]	NO (Exceeds by 16.7x) [cite: 2]	CRITICAL FAIL [cite: 2]
M4 Redesigned Rec.	n log2 n [cite: 2]	1.993 x 10^7 [cite: 2]	0.1993 s [cite: 2]	YES [cite: 2]	PASS [cite: 2]
M5 Sales Analytics	n log2^2 n [cite: 2]	3.972 x 10^8 [cite: 2]	3.972 s [cite: 2]	YES [cite: 2]	PASS [cite: 2]
M5 Report Generator	3n^2 + 10n log2 n + 50 [cite: 2]	3.000199 x 10^12 [cite: 2]	30,001.99 s (~8.33 hrs) [cite: 2]	NO (Exceeds by 50x) [cite: 2]	CRITICAL FAIL [cite: 2]

Explanations & Engineering Action Plan:

• **Failing Modules:** Modules M1, M4 (current design), and M5 (report generator) fail the 10-minute nightly processing window at n = 1,000,000 [cite: 2].

• **Redesign Solutions:**

1. Module M1: Replace nested loops with Hash-Based Deduplication (O(n) time, 0.01 s) [cite: 2].
2. Module M4: Adopt the proposed non-replicated design T(n) = 2T(n/2) + n (O(n log n) time, 0.1993 s) [cite: 2].
3. Module M5: Replace full table scans with Incremental Pre-aggregation / Cumulative Summary tables (< 0.01 s) [cite: 2].

2. Extended Architectural Discussion & Space-Time Trade-offs

In large-scale e-commerce platforms like ShopSphere, algorithmic scalability is governed by space-time trade-offs. While optimizing execution time often requires additional memory structures (such as hash tables for M1 or pre-aggregated summary tables for M5), modern cloud architectures easily absorb linear space overheads ($O(n)$) to achieve exponential time reductions [cite: 1, 2]. • **Hash Collision Handling in M1 Redesign:** Implementing hash-based duplicate detection requires a robust hash function combining product title and seller ID strings. Chaining or open addressing resolves collisions while maintaining expected $O(1)$ lookup and insertion time per product, resulting in overall $O(n)$ time complexity [cite: 1, 2].

• **Incremental Pre-aggregation in M5:** Shifting from batch nightly scans ($3n^2$) to real-time event-driven streaming aggregates ensures that sales analytics and report generation complete instantly upon request without risking server timeouts [cite: 2].

Conclusion: Adopting these algorithmic recommendations guarantees that ShopSphere will maintain high performance and reliability as its product catalogue expands toward 1,000,000 items [cite: 1, 2].

CODE OF CONDUCT & DECLARATION

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action [cite: 2].

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Date: October 12, 2026 [cite: 2] **Signature of the Student:** _____